Crouzeix: Folie à Deux

Crouzeix: Folie à Deux

Nic Berkopec

October 2026

bnic101.github.io/crouzeix-folie-a-deux

Two proofs, July 2026. Same method. Both credit ChatGPT.

Crouzeix: Folie à Deux

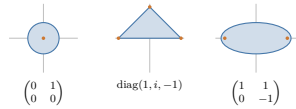
Numerical range

Numerical range

Def. Let $\mathcal{H}$ be a complex Hilbert space and $T \in \mathcal{B}(\mathcal{H})$ a bounded operator. The numerical range of T is

$$W(T) = \{\langle Tx, x\rangle : \ x \in \mathcal{H}, \ \|x\| = 1\}.$$

In finite dimension closed and convex (Toeplitz–Hausdorff). In infinite dimension convex, $\overline{W(T)} \supseteq \operatorname{conv} \sigma(T)$.



Orange: $\sigma(T)$. Blue: $W(T)$.

$\langle Nx, x\rangle = x_2\bar{x}_1$. Normal: hull of spectrum. 2×2: ellipse, foci at eigenvalues, minor axis $|t_{12}|$. Website: more examples, $\|T\|/w(T)$.

Crouzeix: Folie à Deux

Spectral sets

Spectral sets

Def. A set $S \subseteq \mathbb{C}$ is called a K-spectral set for an operator $T \in \mathcal{B}(\mathcal{H})$ if

$$\|p(T)\| \leq K \sup_{z \in S} |p(z)| \qquad \forall\, p \in \mathbb{C}[z].$$

(S bounded, polynomially convex.)

von Neumann's inequality. The disc is a spectral set for any contraction:

$$\|p(T)\| \leq \sup_{z \in \mathbb{D}} |p(z)|, \qquad \|T\| \leq 1.$$

(Also the closed half plane if $\operatorname{Re} T = \frac{T+T^*}{2} \geq 0$.)

von Neumann	Sz.-Nagy
function theory	dilation theory
Blaschke products	unitary dilation of T

Two proofs of von Neumann. Same split for Crouzeix.

Crouzeix: Folie à Deux

Crouzeix conjecture

Crouzeix conjecture

Conjecture (Crouzeix 2004)

Let T be a bounded operator on a Hilbert space $\mathcal{H}$. The numerical range $\overline{W(T)}$ is a 2-spectral set for T.

Remark. This conjectured bound is sharp:

$$W\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = D(0, \tfrac{1}{2}), \qquad p(z) = z,$$

$$\left\| \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \right\| = 1 = 2 \cdot \tfrac{1}{2} = 2 \sup_{D(0,1/2)} |p(z)|.$$

Antecedents.

- von Neumann, only for contractions.
- Okubo–Ando 1975, Berger–Stampfli 1967: the disc is a 2-spectral set.

Berger: $w(T) \leq 1$ iff $T^n = 2V^*U^nV$. Crouzeix 2004: 2×2 case, equality only for disks.

Crouzeix: Folie à Deux

Constants

Constants

Delyon–Delyon	**1999**	$C_\Omega < \infty$	
Crouzeix	**2004**	2	**only for** 2×2
Crouzeix	**2007**	11.08	
Crouzeix–Palencia	**2017**	$1 + \sqrt{2}$	**completely bounded**
Jin	**July 2026**	2	
Lorist–Schwenninger	**Aug 2026**	2	

Technique. The double layer potential.

$C_\Omega = \sup \|f(T)\|$ over $W(T) \subset \Omega$, $\|f\| \leq 1$. Shrink Ω to $\overline{W(T)}$.

Crouzeix: Folie à Deux

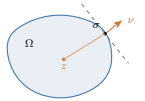
Double layer potential

Double layer potential

Let $\Omega \subset \mathbb{C}$ be an open bounded convex set with smooth boundary $\partial\Omega$ and $\overline{W(T)} \subset \Omega$. $A(\Omega)$: holomorphic on Ω, continuous on $\overline{\Omega}$, sup norm. $C(\partial\Omega)$: continuous on $\partial\Omega$, sup norm. For $\sigma \in \partial\Omega$ denote the outward pointing normal by $\nu = \nu(\sigma)$, and if $z \in \Omega$ define the double layer potential by

$$\mu(\sigma, z) = \frac{1}{2\pi}\Big(\underbrace{\frac{\nu}{\sigma - z}}_{\text{first layer}} + \underbrace{\frac{\bar{\nu}}{\bar{\sigma} - \bar{z}}}_{\text{second layer}} \Big)$$

$= \frac{1}{\pi} \operatorname{Re} \frac{\nu}{\sigma - z}$. First layer: the Cauchy kernel.



- $\mu\, ds = \frac{1}{\pi}\, d\arg(\sigma - z)$
- $\mu \geq 0$
- $\int_{\partial\Omega} \mu\, ds = 2$

Why use? The Cauchy layer leads to a bound on $f(z)$. If we can control the conjugate we win. (Adjoint when applying functional calculus.)

Angle seen from z. Convex: z inside every tangent line. Mass 2 inside, 1 on boundary. Website: drag z.

Crouzeix: Folie à Deux

Double layer potential, cont.

$\mu(\sigma, T) = \frac{1}{\pi} \operatorname{Re}[\nu(\sigma - T)^{-1}] \geq 0$: $\langle Tx, x\rangle \in \Omega$, conjugate by resolvent. Mass $2I$: Cauchy. Lemma: Cauchy–Schwarz.

Double layer potential, cont.

Let

$$C(f,T) = \frac{1}{2\pi i} \int_{\partial\Omega} f(\sigma)(\sigma - T)^{-1}\, d\sigma \in \mathcal{B}(\mathcal{H})$$

and

$$S(f,T) = C(f,T) + C(\bar{f},T)^* \in \mathcal{B}(\mathcal{H}).$$

By definition S is the function calculus integration against the double layer potential $\mu(\sigma, z)$, i.e.

$$S(f,T) = \int_{\partial\Omega} f(\sigma)\, \mu(\sigma,T)\, ds.$$

Lemma (Crouzeix–Palencia 2017)

For $f \in C(\partial\Omega)$, $\|S(f,T)\| = \|f(T) + C(\bar{f},T)^*\| \leq 2\|f\|_{\partial\Omega}$.

So the question we must answer is: how does $\|S(f,T)\|$ control $\|f(T)\|$? In particular, can we estimate $C(\bar{f},T)^*$ or make it vanish all together?

Crouzeix: Folie à Deux

Positivity and the double layer formula

Website: $f(z)$ plus companion as vectors. Extremal f is a Blaschke product (Crouzeix 2004).

Positivity and the double layer formula

Consider

$$2\Phi(f) := f(T) + C(\tilde{f}, T)^* = f(T) + \alpha(f)(T)^*,$$

where

$$\alpha(f)(z) = \frac{1}{2\pi i} \int_{\partial\Omega} \frac{\overline{f(\sigma)}}{\sigma - z}\, d\sigma.$$

Note that $\alpha : A(\Omega) \to A(\Omega)$ is antilinear and $\alpha(1) = 1$.

Positivity of Φ says $2\|\Phi(f)\| \le 2\|f\|_{\partial\Omega}$: the result from the previous slide, found in Crouzeix–Palencia.

Scalar case: $2\Phi(f)(z) = f(z) + \overline{\alpha(f)(z)}$. Disc: $\alpha(f) = \overline{f(0)}$.

An alternate approach (not taken by either ChatGPT proof) is to "hack" the formula by studying extremal functions: f extremal, $\|f(T)\| \le \|2\Phi(f)\|$, which proves Crouzeix. (Bickel, Gorkin, Greenbaum, Overton.)

Crouzeix: Folie à Deux

Crouzeix–Palencia, $1+\sqrt{2}$

Insert $T^* = (T + g(A)^*)^* - g(A)$ in the middle. $f(T) = \begin{pmatrix} f(1) & f(1)-f(0) \\ 0 & f(0) \end{pmatrix}$;

$h(T) = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$. Not a counterexample: $W(T) \not\subset \Omega$.

Crouzeix–Palencia, $1+\sqrt{2}$

$K = C_\Omega$, $\|f\| \le 1$, $T = f(A)$, $g = \alpha(f)$, $\|g\| \le \|f\|$ (Ω convex).

$$TT^*TT^* = T\left(T + g(A)^*\right)^* TT^* - (fgf)(A)\,T^*$$

$$K^4 \le 2K^3 + K^2, \qquad K \le 1+\sqrt{2}.$$

(Ransford–Schwenninger)

Uses only $\|f(A) + g(A)^*\| \le 2$ and $\|g\| \le \|f\|$. Sharp under these (Ransford–Schwenninger 2018):

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \Omega = D(0,\tfrac{1}{4}) \cup D(1,\tfrac{1}{4}),$$

$g = -f(0)$, $-f(1)$ on the two discs; $h = \mp 1$: $\|h(T)\| = 1+\sqrt{2}$.

Here $g(1) = -1$. The Cauchy transform: $\alpha(1) = 1$.

Crouzeix: Folie à Deux

Equivalent formulation

Theorem (Twisted real part)

Let $\mathcal{A}$ be a unital commutative operator algebra, let $\theta : \mathcal{A} \to \mathcal{B}(\mathcal{H})$ be a bounded homomorphism and let $\alpha : \mathcal{A} \to \mathcal{A}$ be a unital antilinear map such that the unital linear map

$$\theta_\alpha : \mathcal{A} \to \mathcal{B}(\mathcal{H}), \qquad a \mapsto \tfrac{1}{2}(\theta(a) + \theta(\alpha(a))^*)$$

is contractive. Then $\|\theta\| \leq 2$.

- Ransford–Schwenninger (2018) show that the twisted real part theorem ⇒ Crouzeix.
- Michael Hartz (2026) proves the theorem using Jin's ChatGPT solution.

$\theta(f) = f(T)$, $\alpha =$ Cauchy transform, $\theta_\alpha = \Phi$. Unital contractive ⇒ positive.

Crouzeix: Folie à Deux

Crouzeix: Folie à Deux

Function theory

$\mathrm{Re}(I - zT)^{-1} \geq 0$ iff $w(T) \leq 1$ (class C_2). Transfer $\Rightarrow$ TRP: $c_z = (1+za)(1-za)^{-1}$, $H(z) = \theta_\alpha(c_z)$, $\theta(c_z) = 2(I - zT)^{-1} - I$, defect $= \frac{1}{2}(\theta(\alpha(c_z))^* - I)$. Concretely $\Delta(w) = \sum_{m \geq 1} w^m \alpha(f^m)(T)^*$.

Function theory

Jin's GitHub; unpublished notes by Hartz.

Theorem (Holomorphic $\to$ resolvent transfer)

Let $T \in \mathcal{B}(\mathcal{H})$ with $\sigma(T) \subset \overline{\mathbb{D}}$. Let $H : \mathbb{D} \to \mathcal{B}(\mathcal{H})$ be a holomorphic function such that $H(0) = I$ and $\mathrm{Re}\, H(z) \geq 0$ for all $z \in \mathbb{D}$. Assume that

$$H(z) - (I - zT)^{-1}$$

commutes with T^* for all $z \in \mathbb{D}$. Then $\|T\| \leq 2$.

- Hartz shows that holomorphic $\to$ resolvent transfer $\Rightarrow$ twisted real part.
- $H(z) - (I - zT)^{-1} = 0$: Okubo–Ando.

Crouzeix: Folie à Deux

Proof

Proof

$\Delta(z) := H(z) - (I - zT)^{-1}$, $R(z) := (I - zT)^{-1}$. On $H^2 \otimes \mathcal{H}$: $\operatorname{Re} M_H \geq 0$, $M_H = M_R + M_\Delta$.

$N_W := \{F = \sum F_n z^n : \ \sum W^n F_n = 0\}$, $\sigma(W) \subset \mathbb{D}$.

- $\Delta(z)W = W\Delta(z) \Rightarrow M_\Delta N_W \subset N_W$.
- $\Delta(0) = 0$, Γ the kernel of N_W at $0 \Rightarrow \Gamma^* M_\Delta \Gamma = 0$.

$W = \frac{1}{2}T^*$:

$$\Gamma^*(M_R + M_R^*)\Gamma = \Gamma^*(M_H + M_H^*)\Gamma \geq 0.$$

$P = \sum 4^{-n} T^{*n} T^n$, $Q = \sum 2^{-n} T^{*n} T^n$, $Y = Q - P \geq \frac{1}{4} T^* T$:

$4Y - YP - PY \geq 0 \ \Rightarrow \ P \leq 2I \ \Rightarrow \ T^*T \leq T^*PT = 4(P - I) \leq 4I.$

Jin: sample the Herglotz kernel of H at $\lambda_i/2$ and at 0.

Compress to functions vanishing at the operator point W. Defect removed exactly. Appendix: computation of Γ, eigenvector argument, Stein identity $P = I + \frac{1}{4} T^* P T$.

Crouzeix: Folie à Deux

Crouzeix: Folie à Deux

Dilation theory

Dilation theory

Lemma (Lorist–Schwenninger, ChatGPT)

Let $\mathcal{H}$ be a Hilbert space and $T \in \mathcal{B}(\mathcal{H})$. Suppose there exists a Hilbert space $\mathcal{K}$, a contraction $Q \in \mathcal{B}(\mathcal{K})$ and an isometry $V \in \mathcal{B}(\mathcal{H}, \mathcal{K})$ such that the operators

$$E_n := 2V^* Q^{*n} V - T^{*n}, \qquad n \in \mathbb{N},$$

are uniformly bounded and commute with T. Then $\|T\| \leq 2$.

- $E_n = 0$: 2-dilation of T^*, $\|T\| \leq 2$.
- $\mathcal{K} = L^2(\partial\Omega, ds; \mathcal{H})$, $Vx = \frac{1}{\sqrt{2}}\mu(\cdot, T)^{1/2}x$, $Q = M_f$, $\|f\|_\infty = 1$, $T = f(A)$:

  $$V^* Q^n V = \Phi(f^n), \qquad E_n = 2\Phi(f^n)^* - f(A)^{*n} = \alpha(f^n)(A).$$

- E_n commutes with $f(A)$; $\|E_n\| \leq \|\theta\|\|\alpha\|$.

Stinespring for Φ. V isometry since $\int \mu = 2I$. $\|\alpha\| \leq 1$ not used. Abstract: θ_α completely positive $\Rightarrow \|\theta\| \leq 2$.

Crouzeix: Folie à Deux

Proof

u: failure of VTx to be a κ-eigenvector of Q^*. $m_1 = \mathrm{Re}\langle (f\alpha(f))(A)x, x\rangle$.

Appendix: both estimates.

Proof

$\kappa := \|T\| > 1$, $T^*Tx = \kappa^2 x$, $\|x\| = 1$ (ultrapower).

$$m_n := \mathrm{Re}\langle E_n T^n x, x\rangle, \qquad u := Q^*VTx - \kappa Vx.$$

$$\|u\|^2 \le 2\kappa^2 - \kappa^3 - \kappa m_1 \quad (n=1)$$

$$\kappa m_1 \ge -\frac{\|u\|^2}{(\kappa-1)^2} \quad \text{(telescope } \kappa m_n - m_{n+1}, \text{ all } n)$$

$$\|u\|^2(1-(\kappa-1)^{-2}) \le \kappa^2(2-\kappa).$$

First line alone: $\kappa \le 1 + \sqrt{1-m_1}$.
$\|\alpha\| \le 1 \Rightarrow m_1 \ge -1 \Rightarrow 1+\sqrt{2}$. $m_1 < 0$ for some 3×3.

Crouzeix: Folie à Deux

Un and Deux

Size of the companion vs. where it lives.

Un and Deux

	Un	Deux
positivity	$\operatorname{Re} H \geq 0$	$\Phi = V^* \pi(\cdot) V$
all powers	$H(w) = I + 2\sum_m w^m \Phi(f^m)$	E_n, all n
companion	$\sum_m w^m \alpha(f^m)(T)^*$	$\alpha(f^n)(T)$
used	commutes with T^*	commutes with T
removal	$\Gamma^* M_\Delta \Gamma = 0$	telescoping

Crouzeix–Palencia: $\|\alpha(f)\| \leq \|f\|$. Un, Deux: $\alpha(f^n)(T) \in \{g(T)\}$.

Crouzeix: Folie à Deux

The canary

Read the disclosures. Humans wrote and checked the proofs.

The canary

- Jin: ChatGPT proposed sampling at $\lambda_i/2$ and at 0.
- Lorist–Schwenninger: GPT-5.6, 25–26 July. $\Phi(f^{n+1})^* f(A)x - \|f(A)\|\Phi(f^n)^* x$ controlled uniformly in n. Saw Jin 28 July.
- Hartz: ChatGPT to discuss Jin.

Same step twice: one $f \to$ all f^n.

Crouzeix: Folie à Deux

Completely bounded

Paulsen: cb-norm $\leq 2 \Rightarrow$ similar to an operator with $\overline{W(T)}$ a complete spectral set. Fill in the reference.

Completely bounded

Conjecture

For every $m \geq 1$ and $F = [f_{ij}] \in M_m(A(\Omega))$, $\|F(T)\| \leq 2 \sup_{z \in \Omega} \|F(z)\|$.

- Known: $1 + \sqrt{2}$ (Crouzeix–Palencia); 2 for discs, weighted shifts (Crouzeix–Greenbaum 2025).
- $M_m(A(\Omega))$ not commutative. Un, Deux need the companion to commute with $f(T)$.
- Crouzeix–Palencia: only $\|\alpha\| \leq 1$.
- Last week: [reference]. OpenAI SlopDrop (Burnstein, Iosevich, Krause, Yavicoli).

Crouzeix: Folie à Deux

References

References

- S. Jin, *The numerical range is a 2-spectral set*, Preprints.org 202607.1919, 27 July 2026.
- M. Hartz, *Notes on the solution of Crouzeix's conjecture*, 31 July 2026.
- E. Lorist, F. L. Schwenninger, *A solution to Crouzeix's conjecture*, arXiv:2608.03841.
- M. Crouzeix, *Bounds for analytical functions of matrices*, Integral Equations Operator Theory 48 (2004).
- M. Crouzeix, C. Palencia, *The numerical range is a* $(1+\sqrt{2})$*-spectral set*, SIAM J. Matrix Anal. Appl. 38 (2017).
- T. Ransford, F. L. Schwenninger, *Remarks on the Crouzeix–Palencia proof*, SIAM J. Matrix Anal. Appl. 39 (2018).
- R. Clouâtre, M. Ostermann, T. Ransford, *An abstract approach to the Crouzeix conjecture*, J. Operator Theory 90 (2023).
- K. Bickel, P. Gorkin, *Blaschke products, level sets, and Crouzeix's conjecture*, J. Anal. Math. 152 (2024).
- B. Delyon, F. Delyon, *Generalization of von Neumann's spectral sets*, Bull. Soc. Math. France 127 (1999).

`bnic101.github.io/crouzeix-folie-a-deux`